climatekit

Unified climate indices in R

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The question

How should we compute the standard suite of climate indices from daily observations in a single reproducible R interface?

Thirty indices, five families, one consistent tidy output schema.

Why it matters

- **National statistical offices** publish annual climate-index tables
- **Agricultural research** tracks growing-season and heliothermal indices
- **Insurance underwriters** price parametric drought and heat products
- **Public-health authorities** monitor heat-index exceedances in real time

What is already out there

- **`SPEI`**: drought indices only, narrow scope[1]
- **`climdex.pcic`**: standard ETCCDI indices, archived on CRAN
- **`Evapotranspiration`, various agri packages**: fragmented, inconsistent output schemas
- **Station-level tooling**: bespoke scripts, no reusable public interface

The gap: **no single package unifies the standard workflow with a tidy output schema**.

[1] Begueria & Vicente-Serrano (2017), *SPEI: Calculation of the Standardised Precipitation-Evapotranspiration Index*, R package v1.7.

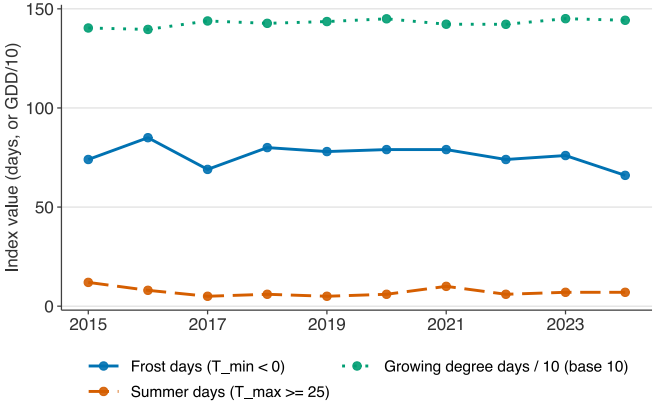
What climatekit offers

1. **Coverage**: 30+ indices across 5 families (temperature, precipitation, drought, agroclimatology, thermal comfort)
2. **Interface**: every function returns the same four-column tidy schema; outputs compose via `rbind()`
3. **Provenance**: pure computation, no API calls, no network dependency, 133 unit tests

On CRAN since March 2026. Definitions follow ETCCDI recommendations[2].

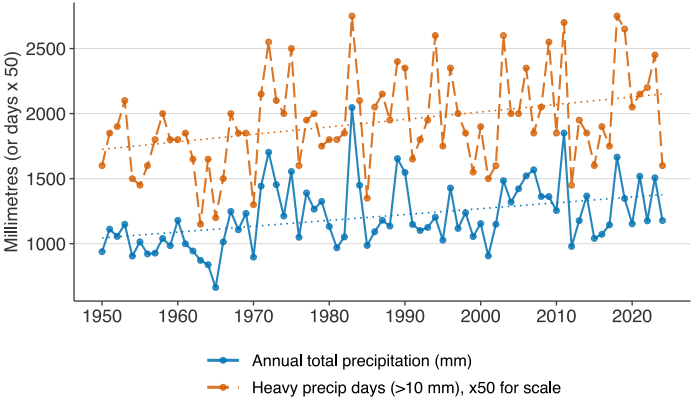
[2] Zhang et al. (2011), *Indices for monitoring changes in extremes based on daily temperature and precipitation data*, WIREs Climate Change 2(6).

Temperature: frost days, summer days, GDD



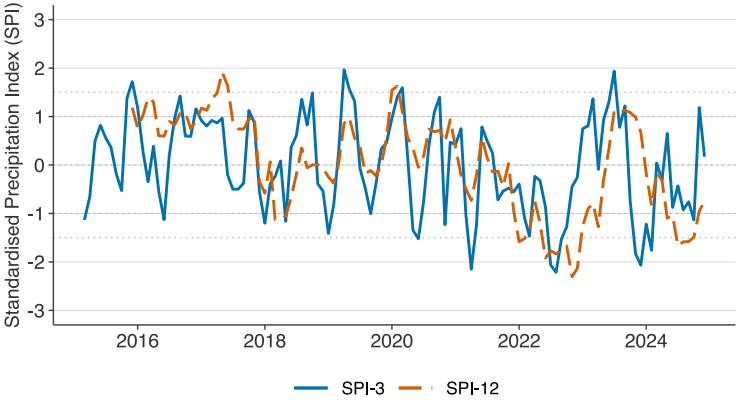
Annual temperature indices, Central Park.

Precipitation: totals, intensity, extremes



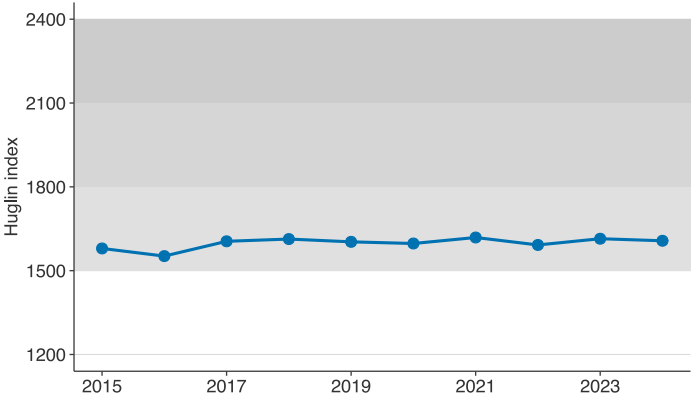
Annual totals (blue) and heavy-precip days above 10 mm (orange, scaled 50x), Central Park.

Drought: standardised precipitation indices



SPI-3 and SPI-12, Central Park. Dashed lines at ± 1 mark moderate drought thresholds.

Agroclimatology: Huglin heliothermal index



Huglin index, synthetic 10-year temperate site at $51.5°$N. Shaded bands mark viticultural suitability.

Central formulas

Frost days (count below 0°C):

$$\mathrm{FD} = \sum_t \mathbf{1}\{T_{\min,t} < 0^\circ\mathrm{C}\} \tag{1}$$

Growing degree days (cumulative heat above base b):

$$\mathrm{GDD}_b = \sum_t \max\left(T_{\mathrm{mean},t} - b,\ 0\right) \tag{2}$$

Standardised Precipitation Index (McKee et al. 1993):

$$\mathrm{SPI}_s = \Phi^{-1}[F(P_s;\ \alpha, \beta)] \tag{3}$$

Package at a glance

Function families:

- **Temperature**: 8 functions
- **Precipitation**: 7 functions
- **Drought** (SPI, SPEI, PET): 3 functions
- **Agroclimatology** (Huglin, Winkler, Branas): 5 functions
- **Thermal comfort**: 4 functions
- **Dispatchers**: `ck_compute()`, `ck_available()`, `ck_metadata()`

Deps: `cli`, `stats`, `tools`. R $\geq$ 4.1.0.

Uniform output

Every function returns:
`period` | `value` | `index` | `unit`
Outputs compose via `rbind()` into panel data.

Minimal working example

```r
library(climatekit)

# Three indices from Central Park daily observations
fd   <- ck_frost_days(tmin,  dates, period = "annual")
gdd  <- ck_gdd(tmin, tmax,   dates, base = 10, period = "annual")
spi3 <- ck_spi(precip,       dates, scale = 3)

# Compose into a panel for plotting or modelling
panel <- rbind(fd, gdd, spi3)
```

All three calls share the same four-column schema, so `rbind()` works without reshaping or renaming.

Central Park temperature trends, 1950 to 2024

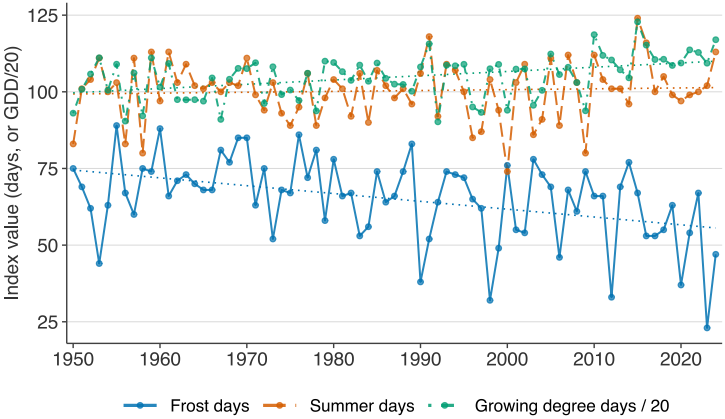
Data. NOAA GHCND station USW00094728. 75 years of daily $T_{\min}$, $T_{\max}$, and precipitation. Complete record through 2024.

Question. *How have temperature indices evolved in the US Northeast over seven decades?*

Why this case. Flagship US long-record station. Northeast warming debate active in *J. Climate*[3]. Three indices with distinct signals (warming, stable, trending). Reproducible with three public function calls.

[3] Brown & DeGaetano (2013), *Trends in U.S. surface humidity*, J. Applied Meteorology 52.

Central Park: three temperature indices, 1950 to 2024



Frost days: -2.6 *per decade. Summer days:* $+0.3$ *per decade. GDD (base* $10°\mathrm{C}$*,* $\div 20$*):* $+28$ *per decade. Source: NOAA GHCND station USW00094728.*

What climatekit does not yet do

- **Daily observations only**: no sub-hourly heat-wave indices
- **No gridded input helpers**: users reduce rasters to per-cell series first
- **SPI / SPEI use gamma distribution**: Pearson III not exposed for arid regions
- **No station homogenisation**: quality control is upstream user responsibility

v0.2.0 roadmap: hourly heat-wave indices, gridded input wrappers, additional fire-weather variants (Canadian FFMC, DC, ISI).

Contact, code, paper

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